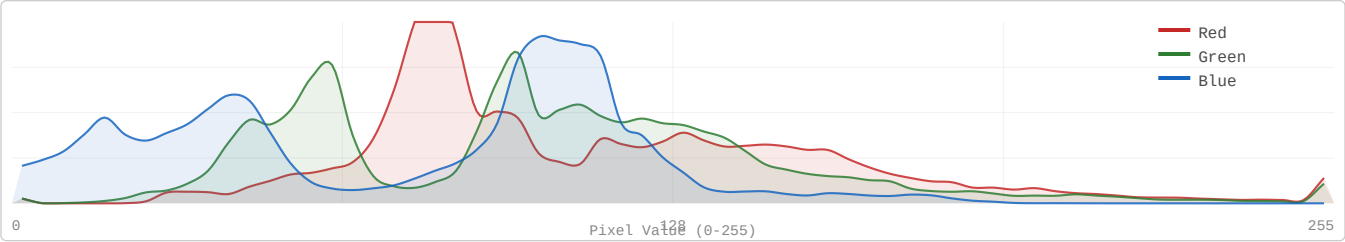
Decorrelation Gap: KODIM03

768x512 | 24-bit RGB | PNG (lossless)

Boy and Girl in Field | Three-dimensional (PC1 = 69.27%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.7185	0.2890
G	0.7185	1.0000	0.5534
B	0.2890	0.5534	1.0000

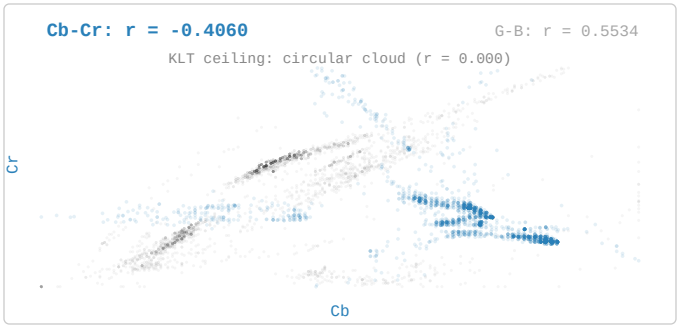
Cb-Cr: -0.406

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.3883	-0.0925
Cb	-0.3883	1.0000	-0.4060
Cr	-0.0925	-0.4060	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.718536	-0.388305	0.3883
R-B → Y-Cr	0.288997	-0.092542	0.0925
G-B → Cb-Cr	0.553408	-0.406037	0.4060
Avg |r|	0.520314	0.295628	0.2956

Decorrelation Gap: KODIM03

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Boy and Girl in Field | Three-dimensional (PC1 = 69.27%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

43.2% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

52.0%

0% Fully predictable — 100% Fully independent

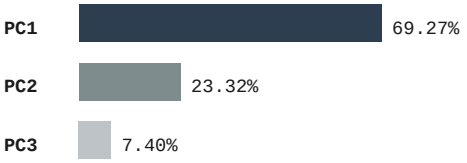
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	44.28 dB
G	48.16 dB
B	41.78 dB
Overall	44.02 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5843	-0.6635	-0.4672	3974.75
PC2	-0.5749	-0.0679	0.8154	1338.29
PC3	-0.5728	0.7451	-0.3418	424.76

8. PCA Derived Metrics

Condition Number

9.36

PC1/PC2: 2.97 | PC2/PC3: 3.15

Blue Channel Independence

52.0%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.66)

9. Redundancy Profile

Classification: Three-dimensional | CN: 9.36 | PC1: 69.3%

RGB Correlation: Avg |r| = 0.520314 | R-G: 0.7185 | R-B: 0.2890 | G-B: 0.5534

YCbCr Residual: Avg |r| = 0.295628 | Y-Cb: -0.3883 | Y-Cr: -0.0925 | Cb-Cr: -0.4060

Efficiency: 43.2% of KLT ceiling | Gap: 0.2956 avg |r| remaining

Blue Channel: 52.0% independent | 4:2:0 PSNR: 44.02 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.